

ABNORMAL SOUND DETECTION BASED ON DOMAIN GENERALIZATION

Technical Report

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ABSTRACT

This technical report describes the anomalous sound detection system we submitted for DCASE 2025 Task 2. DCASE Task2 aims to solve the core pain points of machine health monitoring in industrial scenarios. Compared to last year's task, this year's additional datasets have been added, and small and difficult to detect domain shifts have been added, but the main focus is still on the unsupervised learning framework to achieve the first detection of unknown abnormal sounds from industrial machines, while overcoming the interference of domain shifts. At present, the pain points of this task are: 1. domain shift sensitivity, 2. finite sample learning bottleneck, and 3. complexity of abnormal patterns. In order to solve these problems, we submit four methods for detecting abnormal sounds in machine status. In the first and second methods, we used the joint model composed of VQ-VAE and PixelSNAIL for anomaly detection, and in the third and fourth methods, we used AE and flow models for anomaly detection. All four methods use feature vectors extracted from convolutional neural networks to identify anomalous sounds through anomaly detection algorithms. Experiments on the development set show that the performance of these four methods is better than that of the benchmark model.

Index Terms— DCASE, anomalous sound detection, domain generalization

1. INTRODUCTION

This report describes the Anomalous Sound Detection (ASD) method developed for DCASE 2025 Challenge Task 2[1]. For Task 2, the task organizer provided a baseline system based on an autoencoder (AE)[2] with two different operating modes. Simple autoencoder mode and selective Mahalanobis mode. This task is mainly set for the first-shot problem under the domain generalization requirement. First, the primary goal of the first problem is to rapidly deploy an ASD system for a new type of machine without the need for machine-specific hyperparameter tuning. Second, there may be a limited number of machines of the machine type, in which case the system should be able to use several machines of the machine type to train the model. Finally, the system must function well when new machine type attribute information is available and not. Unlike previous years, this year supplementary data is provided for each type of machine, which may include clean machine data or only noise data.

In recent years, the Vector Quantization Variational Autoencoder (VQ-VAE)[3] has demonstrated significant advantages in the field of cross-domain feature reconstruction due to its strong representation capability in the discrete latent space and its feature decoupling characteristic: its quantization codebook can capture the essential structural information of machine sounds, effectively

improving the discrimination of reconstruction errors for abnormal samples. At the same time, the Normalized Flow (Flow) [4] model, with its precise and reversible probability density estimation capability, provides theoretical support for the modeling of complex acoustic distributions and likelihood calculations, and can overcome the limitations of traditional generative models in assuming the distribution of the latent space. Based on these advantages, this paper uses the frameworks of VQ-VAE and Flow for anomaly detection. It is particularly important to note that the proposed method in this work is divided into four categories: the first and second methods are based on VQ-VAE, and the third and fourth methods are based on the Flow model.

2. EXPERIMENT

2.1. Experiment settings

Experiments were performed on a system using Ubuntu version 22.04. The setup was used in conjunction with Python 3.10.14 and PyTorch 2.2.2 during training and testing and was run on an NVIDIA GeForce RTX 3080 GPU.

2.2. Dataset

The DCASE 2025 task 2 dataset provides a domain-oriented generalization benchmark for industrial anomalous sound detection (ASD), and contains three core components: the development dataset (ToyADMOS2[5] and MIMIIDG[6]) covers seven types of machines: fans, gearboxes, bearings, slide rails, toy cars, toy trains, and valves, each of which provides 990 normal audio segments in the source domain, 10 normal audio segments in the target domain, and 100 supplementary data segments (pure machine sound/ environmental noise) and 200 test samples (100 normal + 100 abnormal); The additional training set contains seven types of new machines that are completely different from the development set, and supports the first detection of unknown machines in zero-shot scenarios. The evaluation set contains 200 segments/classes of unlabeled test samples corresponding to the additional training set. The dataset records mono audio (10-12 seconds, signal-to-noise ratio 5-15dB) through industrial field recording, and the system introduces four major field offset factors - machine speed (1800-2400 RPM), environmental noise (three types of factory background), mechanical load and microphone layout (four positions), and is equipped with a metadata file (CSV format) of annotation domain offset parameters, and provides high signal-to-noise ratio pure data (machine sound 50dB/ environmental noise 35dB) to support robust modeling, and its multi-dimensional design significantly improves the generalization proficiency verification validity of industrial anomaly detection under complex working conditions.

2.3. Features Processing

To develop an effective representation for anomalous sound detection from raw audio signals, we employed log-Mel spectral features as the network input. Our experimental configuration utilized a 16 kHz sampling rate with 128 Mel-frequency bins and a frame length of 64 ms. We further constructed 640-dimensional feature vectors by concatenating five consecutive frames, forming the primary input to our model. To improve feature robustness, three augmentation techniques were applied: pitch shifting, clipping distortion, and random time-frequency perturbations.

3. METHOD

3.1. VQ-VAE

In this method, anomaly detection is achieved through bimodal feature extraction, VQ-VAE discrete coding, and PixelSNAIL distribution modeling. Firstly, the metadata features are generated by audio-text contrast learning [7] to generate 512-dimensional vectors. The original audio was converted into a logarithmic spectrum by STFT (window length 1024, jump 512) and a 128 Weimeier filter set, and the features of 5 segments and 128 frames were intercepted. Method 2 introduces the feature of adaptive band filtering optimization spectrum (method 1 is omitted). The fused feature input VQ-VAE is used in the encoder to enhance feature extraction by the residual module with attention gating (AG-Resblock), and the L1 reconstruction loss is trained. VQ-VAE outputs a discrete coding matrix and inputs the PixelSNAIL[8] model to learn the normal sequence distribution. The anomaly score is weighted by two parts: 1. Reconstruction error: the L1 distance between the VQ-VAE decoded output and the original input, 2. Predicted loss: the negative logarithmic likelihood of PixelSNAIL to discrete encoding. The dual-path mechanism comprehensively evaluates the feature reconstruction and potential distribution bias.

3.2. Flow

In order to solve the problem of detection performance degradation caused by cross-domain data imbalance, this study uses the hybrid strategy of SMOTE and Mixup to synergistically enhance the source/target domain data: SMOTE generates synthetic samples that conform to the distribution characteristics of the target domain in the feature space to fill in the sparse data area, and Mixup constructs continuous transition samples through cross-domain convex combinations. The enhanced dataset after probabilistic blending significantly alleviates the domain shift effect and provides a balanced distribution input for subsequent modeling. On this basis, a Normalizing Flow model is deployed to achieve high-precision anomaly detection, which maps inputs to latent space through an invertible transformation chain and calculates accurate log-likelihood values based on the variable transformation theorem. The core advantages of the Flow model are: 1) enhanced data to effectively suppress source domain overfitting and improve the generalization ability of the target domain; 2) Exact density estimation quantifies the degree to which a sample deviates from the normal distribution and generates an interpretability anomaly score; 3) The double-shot structure of the hidden space naturally models the invariant features of complex domains. Experiments have shown that the effect of this method is higher than baseline.

3.3. AE

When the model trained based on a highly skewed distribution (source domain data dominant) migrates to the target domain, the

performance degradation is not only due to the real anomaly pattern difference, but also due to the systematic bias caused by Domain Shift, which is manifested as an increase in the false positive rate of normal samples in the target domain and a surge in the missed detection rate of abnormal samples. In order to solve this bottleneck, we innovatively introduce a dual data augmentation strategy on the autoencoder baseline system: firstly, SMOTE (synthetic minority oversampling technology) is used to intelligently expand the scarce target domain samples in the feature space, and the synthetic data that conforms to the distribution characteristics of the target domain is generated through linear interpolation of k-nearest neighbor samples. At the same time, the Mixup is deployed to construct a convex combination of the source domain and the target domain samples, forcing the model to learn domain invariant feature representations. The newly added synthetic data and the original data are combined to form a domain adaptive training set, which not only alleviates the problem of category imbalance through the synergistic mechanism of feature space interpolation and input space mixing, but also explicitly models the transition state between domains, significantly weakens the performance degradation caused by domain shift, and makes the model obtain a more robust anomaly discrimination ability on the target domain.

4. RESULTS

Table 1 shows the results of Methods 1, 2, 3, and 4 on the development dataset, including AUC and pac for source and target domains. As shown in the table, our results outperformed the baseline results on most machine types.

5. REFERENCES

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Table 1: Anomaly detection results for different machine types

Model	Score	ToyCar	ToyTrain	Bearing	Fan	Gearbox	Slider	Valve
Baseline (Mahala)	s_AUC	63.10	61.99	79.37	81.82	54.43	75.35	55.69
	t_AUC	37.35	39.99	42.70	74.35	51.58	68.11	53.61
	pAUC	51.04	48.21	53.44	55.74	58.82	49.05	51.26
Method1	s_AUC	71.20	63.36	56.16	54.40	54.60	71.80	50.38
	t_AUC	72.42	64.92	61.52	65.04	56.56	51.72	51.08
	pAUC	50.63	53.37	57.58	55.58	53.16	55.58	50.31
Method2	s_AUC	70.36	70.8	59.72	73.44	61.44	70.36	56.88
	t_AUC	75.56	54.28	60.44	48.44	51.88	75.56	52.04
	pAUC	53.68	53.68	52.94	52.10	51.78	54.63	52.21
Method3	s_AUC	70.62	60.76	60.70	70.56	64.48	69.24	61.72
	t_AUC	53.96	55.58	49.34	38.75	50.49	49.28	67.74
	pAUC	49.89	50.69	60.63	49.84	53.10	51.73	57.11
Method4	s_AUC	73.48	61.16	61.53	77.94	72.26	73.81	62.13
	t_AUC	52.12	56.46	56.34	39.15	51.46	49.52	67.18
	pAUC	49.70	50.19	58.12	61.05	50.63	55.05	57.35